Kenji Watanabe

Senior Software Developer — Security and Systems Engineering

Singapore | kenji.watanabe@example.com | linkedin.example.com/in/kenji-watanabe | github.example.com/kwatanabe-sec

PROFESSIONAL SUMMARY

Senior security and systems engineer with 14 years of experience across embedded systems, cloud hosting infrastructure, and digital banking. Combines low-level systems knowledge in C, Rust, and Linux with application and cloud security, including threat modeling, zero-trust architecture, detection engineering, and incident response. Designed a zero-trust service identity model for 260 banking services that removed long-lived database passwords, and cut mean time to detect high-severity threats from nine hours to 40 minutes. Embeds security into engineering workflows through design reviews, automated scanning in CI, and practical guidance rather than gatekeeping. Leads incident response exercises, mentors engineers on secure coding, and is interested in memory-safe systems programming and eBPF-based runtime security.

TECHNICAL SKILLS

Programming languages: Advanced: C, Rust, Go, Python. Working knowledge: C++, Java, Bash, x86-64 assembly (reading).

Application security: Threat modeling (STRIDE), secure code review, OWASP Top 10, SAST and DAST (Semgrep, CodeQL, OWASP ZAP), dependency scanning and SBOMs, fuzzing (libFuzzer, AFL++).

Infrastructure and cloud security: Zero-trust architecture, mTLS, SPIFFE and SPIRE, HashiCorp Vault, AWS security (IAM, KMS, GuardDuty, Security Hub), Kubernetes security, network segmentation.

Systems and Linux: Linux kernel hardening, SELinux, seccomp, eBPF, embedded Linux, secure boot, firmware signing, hardware security module (HSM) integration.

Detection and response: SIEM detection engineering (Elastic Security), Falco, Sigma rules, incident response, tabletop exercises, digital forensics basics.

Applied cryptography: TLS configuration, key management, signing and verification workflows, PKI operations.

DevOps tools: Kubernetes, Terraform, Docker, GitHub Actions, GitLab CI.

Industry experience: Digital banking and payments, cloud hosting, industrial embedded systems; supported engineering controls for banking and payment card security standards.

PROFESSIONAL EXPERIENCE

Straitsgate Digital Bank (digital banking) — Singapore

Senior Security Engineer, Systems and Infrastructure — *Jan 2021 – Present*

- Designed and led the rollout of zero-trust service identity using SPIFFE/SPIRE and mTLS across 260 microservices, with short-lived database credentials issued by HashiCorp Vault; long-lived service passwords were fully removed within 14 months.
- Built a STRIDE-based threat modeling program that is part of every architecture review; the team completes about 70 reviews per year with engineering and product teams.
- Developed detection rules in Elastic Security and Falco runtime policies for Kubernetes, reducing mean time to detect high-severity events from nine hours to 40 minutes in quarterly red-team exercises.
- Rewrote a C-based proxy between payment services and hardware security modules in Rust after fuzzing found memory-safety defects; the new proxy has had no memory-safety findings in continuous fuzzing and kept p99 latency within 1 ms of the original.
- Lead incident response tabletop exercises with engineering, operations, and compliance, and mentor two security engineers on secure design reviews.

Kairos Cloud Hosting (cloud infrastructure hosting) — Tokyo, Japan

Senior Security Engineer (Security Engineer until 2019) — *Apr 2016 – Dec 2020*

- Hardened Linux hypervisor hosts with SELinux, seccomp profiles, and kernel live patching, reducing the time to patch critical vulnerabilities across 4,000 hosts from 21 days to 3 days.
- Integrated Semgrep, OWASP ZAP, and dependency scanning with SBOM generation into CI for 120 repositories, using severity thresholds agreed with engineering leads to avoid alert fatigue.
- Coordinated bug bounty triage for about 140 external reports per year, working with service owners on fixes and disclosure timelines.

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- Built an eBPF-based process execution monitor for hypervisor hosts that detected unexpected binaries with under 2% CPU overhead.

Hoshino Embedded Systems (industrial embedded systems) Osaka, Japan
Systems Engineer *Apr 2012 – Mar 2016*

- Developed embedded Linux firmware in C for industrial controllers, including device drivers and a watchdog-based recovery process.
- Implemented secure boot and signed over-the-air firmware updates, preventing installation of unsigned images on 30,000 field devices.
- Introduced static analysis and code review checklists that reduced defects found in field testing by about 30%.

SELECTED PROJECTS

Fuzzing harness for payment message parsers
Technologies: Rust, C, libFuzzer, AFL++, continuous fuzzing in CI

- **Problem:** Binary message parsers used by payment services had never been fuzzed.
- **Contribution:** Wrote fuzzing harnesses and structure-aware mutators, and set up CI jobs that fuzz continuously.
- **Result:** Found 11 crashes and two memory-safety bugs, all fixed before they were reachable in production.

execwatch — eBPF runtime monitor
Technologies: Rust, eBPF (aya), Linux, Elastic Security

- **Problem:** Container runtime monitoring produced too much noise on build hosts to be useful.
- **Contribution:** Built an eBPF agent that records process executions with container context and filters known build activity.
- **Result:** Alert volume fell by about 70% while the agent still detected simulated intrusions in red-team exercises.

Secrets scanning pre-commit hook
Technologies: Go, pattern and entropy checks, Git hooks

- **Problem:** Credentials occasionally reached shared repositories.
- **Contribution:** Built a fast pre-commit and CI scanner with an allowlist workflow for false positives.
- **Result:** Rolled out to 400 engineers; secrets found in repositories dropped from a monthly occurrence to rare isolated cases.

EDUCATION

Kansai Institute of Technology Osaka, Japan
M.Eng. Information Science *2010 – 2012*

- Thesis: detecting control-flow hijacking in embedded firmware with lightweight instrumentation.

Kansai Institute of Technology Osaka, Japan
B.Eng. Computer Engineering *2006 – 2010*

- Coursework: operating systems, computer architecture, cryptography, computer networks.

HACKATHONS, CONFERENCES, AND PROFESSIONAL ACTIVITIES

- **SecureStack Asia, Singapore (2024)** — Speaker: “Removing passwords from a bank”, on SPIFFE and short-lived credentials in production.
- **Lion City CTF (2023)** — Member of a five-person team in a capture-the-flag competition focused on binary exploitation and web security.
- **Open Source Security Hackathon, Tokyo (2022)** — Built a prototype tool that generates software bills of materials (SBOMs) for embedded firmware images.

CERTIFICATIONS AND ADDITIONAL QUALIFICATIONS

Certifications (illustrative): Offensive Security Practitioner Program, Harborview Security Institute (2019); Cloud Security Architecture Track, Meridian Cloud Academy (2021).

Mentoring: Runs a secure-coding study group for engineers at Straitsgate Digital Bank.

Languages: Japanese (native), English (fluent), Mandarin Chinese (basic).

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